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Alessio Boetti

Date of birth: 07/01/1999

Nationality: Italian

Gender: Male

CONTACT

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ABOUT ME

AI Engineer and Data Scientist with solid experience in designing and implementing predictive models on complex data and in creating data-driven solutions to guide strategic decisions and optimize business performance. Specialized in machine learning, applied statistics, and end-to-end solutions — from exploratory analysis to production deployment — both on cloud and on-premise environments.

WORK EXPERIENCE

Proxima Group Rome (Remote), Italy

Website www.proximagroup.eu

Consultant Analytics Intelligence

06/10/2025 – Current

Contributed to the development of an ERM platform that enables banks to anticipate financial trends, optimize risk management, and support strategic decision-making through advanced forecasting and simulation models.

TAS S.p.A. Casalecchio di Reno, Bologna (Remote), Italy

Website www.tasgroup.eu

Banking Data Scientist & Machine Learning Engineer

31/10/2022 – 03/10/2025

In short: Development and deployment of end-to-end data-driven machine learning models and AI solutions in the financial sector, with a focus on anomaly and fraud detection, time series analysis and forecasting, NLP-based and GenAI-powered chatbots, business process automation, and business intelligence.

Full lifecycle development and management of:

- **4 Fraud Detection projects:** 4 large-scale banking fraud detection solutions using **AWS, Apache Spark & PySpark, Terraform, Docker, Kubernetes**, significantly improving fraud detection accuracy and reducing clients' financial losses (more than €25k of frauds intercepted).

Projects:

- predictive models with monitoring systems and outputs explanations (Explainable AI), reviewed through a formal audit process
 - business intelligence and analytics
 - alert rules AI generation
 - anti-fraud customer screening chatbot
- **Liquidity forecasting system:** treasury liquidity forecasting engine, optimizing clients' financial cash management and predicting liquidity needs through time series analysis/modeling and statistical techniques.
- **Advanced GenAI chatbots:** GenAI-powered chatbots using **LLMs** and **NLP** for B2B solutions, customer support, internal support, and company website integration.
- **Anomaly detection for infrastructure monitoring:** monitoring of internal technical infrastructure through system logs to identify malfunctions and anomalous behavior in real time, reducing downtime and manual overhead.
- **Statistical data analysis & reporting:** detailed business intelligence reports on banking customers demographics and behaviour.

Process optimization: Streamlined and standardized testing cycles and infrastructure processes by **automating manual activities**, drastically reducing operational times.

Scalable & secure testing and deployment with automated infrastructure: used containerization (Docker), orchestration (Kubernetes, Rancher, Argo, Docker Compose), CI/CD and DevOps/MLOps tools and pipelines (Pytest, Git/GitHub/GitLab, Coverity, Black Duck, SonarQube, Kaniko),

access management tools (Keycloak, Duo), and versioned databases (PostgreSQL, Liquibase).

Led an **AI & ML internal training course** to educate senior technical colleagues on the foundations and business value of Artificial Intelligence, covering both theoretical concepts and hands-on Python code demonstrations.

EDUCATION AND TRAINING

22/09/2021 – 03/10/2025 Florence, Italy

● **Master's Degree in Artificial Intelligence (Information Engineering, LM-32)** Università degli Studi di Firenze (University of Florence)

Thesis: Graph Neural Networks for Banking Fraud Detection

AI projects: implemented 2 deep learning models to detect anomalous images, a machine learning model to detect fraudulent banking transactions, multiple deep learning models to correctly classify images, a deep learning model to resist adversarial attacks on images, built and trained an LLM from scratch.

Statistics, Generative AI, Deep Learning, Neural Networks, Machine Learning, Natural Language Processing (NLP), LLMs, Computer Vision, Adversarial Learning, Anomaly Detection, Out-of-Distribution Detection, Big Data, Innovation Management, Network Security, Robotics.

Website <https://www.unifi.it/it> | **Level in EQF** EQF level 7

09/2022 – 11/2022 Marghera, Venice (Remote), Italy

● **Academy "Banking Data Scientist"** Umana Forma S.r.l & UMANA S.p.A.

Course on AWS, R, Python, Data Science, Big Data, Apache Spark & PySpark, SQL & NoSQL Databases. Final project on Banking Fraud Detection using Machine Learning models.

09/2017 – 05/2021 Florence, Italy

● **Bachelor's Degree in Electronics and Telecommunications Engineering (Telecommunications course, Information Engineering, L-8)** Università degli Studi di Firenze (University of Florence)

Thesis: Design of a congestion control protocol in TCP/IP networks through machine learning algorithms

Mathematics, Physics, Electronics, Electrical Engineering, Telecommunications

Website <https://www.unifi.it/it> | **Level in EQF** EQF level 6

2017 Sesto Fiorentino (Florence), Italy

● **High School Diploma (Secondary Education Certificate)** I.S.S. – Liceo Scientifico A. M. Enriques Agnoletti

Level in EQF EQF level 4

LANGUAGE SKILLS

MOTHER TONGUE(S): Italian

Other language(s):

English

Listening C1

Spoken production B2

Reading C1

Spoken interaction B2

Writing C1

French

Listening A1

Spoken production A1

Reading A1

Spoken interaction A1

Writing A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Digital Marketing | SEO | WordPress | ECDL | Google Analytics | Social Network | Email Marketing | C++
| Python | R | Microsoft Office

SOFT SKILLS

● **Management**

Ability to manage stress, time, and priorities.

● **Leadership**

Coordination, teamwork, leadership, decision-making, problem-solving, and team management skills developed as a team leader on projects at TAS S.p.A.

● **Public Speaking**

Public speaking skills developed through a theater course.

TECHNICAL SKILLS

● **Hard Skills and Stacks**

Areas of expertise: Statistics, Machine Learning, Deep Learning, Tabular Data, Graph Data, Graph Neural Networks, Time Series Analysis, Natural Language Processing (NLP), Generative AI (LLM, RAG, Fine-tuning, Agents, Embeddings, Vector Databases), Computer Vision, Anomaly Detection, Adversarial Learning, Supervised/Unsupervised/Self-Supervised Learning, Explainable AI (XAI), Transformers, CNN, RNN, LSTM, Serverless Architectures, Microservices Architectures.

Languages: Python, R, SQL

Libraries:

- Data Analysis: Pandas, NumPy, Matplotlib, Plotly, Seaborn, etc.
- Machine Learning: Scikit-Learn, XGBoost, CatBoost, Statsmodels, Nixtla
- AI & GenAI: PyTorch, LangChain, OpenAI SDK
- Graphs: PyTorch Geometric, Deep Graph Library, NetworkX
- Optimization: HyperOpt, Optuna

Cloud & Distributed computing: AWS (DynamoDB, SageMaker, S3, Lambda, Glue, QuickSight, API Gateway, etc.), Azure, Apache Spark & PySpark

Containerization: Docker

Orchestration: Docker Compose, Kubernetes, Rancher, Argo

CI/CD, MLOps, DevOps, IaC: Git/GitHub/GitLab, CI/CD pipelines, Terraform, Keycloak & Duo, Azure DevOps

Databases: PostgreSQL, MongoDB, SQLAlchemy, Liquibase

CERTIFICATIONS

- **AWS Certified Machine Learning Specialty, 2023, Udemy**
 - **Customer Analytics in Python, 2023, Udemy**
 - **Introduction to Git for GitLab projects, 2023, Udemy**
 - **ISO 56000 Innovation Management, 2021, University of Florence**
 - **B2 Cambridge English, 2016, British Institute of Florence**
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Sesto Fiorentino (Florence), 05/01/2026



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